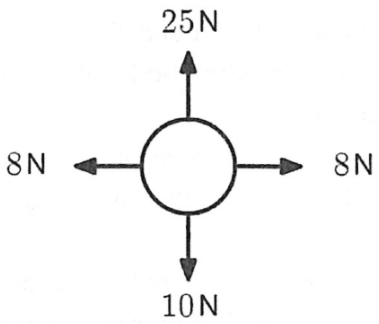
Forces: Equilibrium

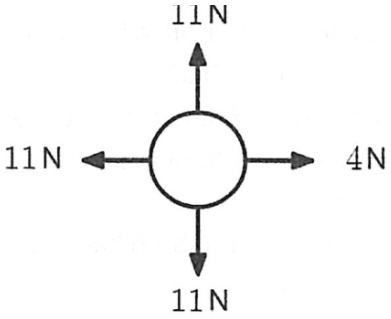
Questions 1–11

Question 1. Find the resultant force in each of the following cases, stating both the value and direction:

a)

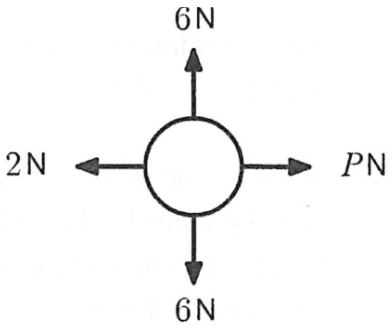


b)

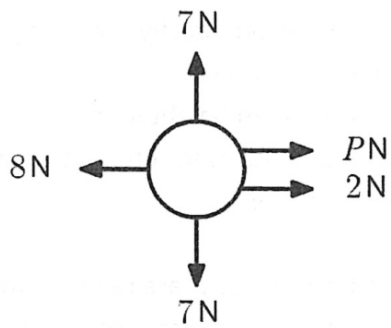


Question 2. Given that each of these particles is stationary, find the value of P:

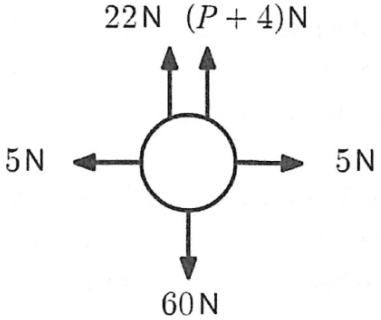
a)



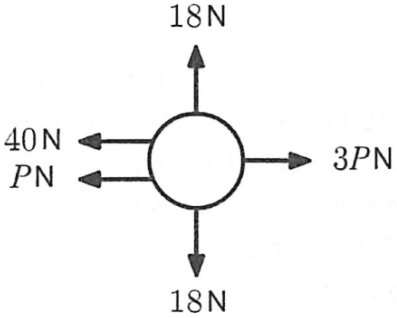
b)



c)

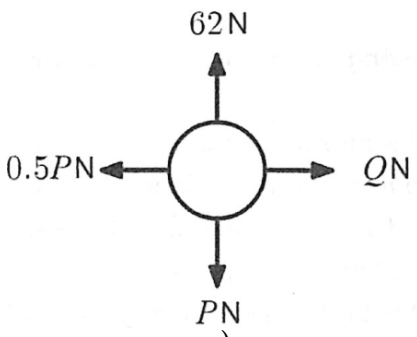


d)

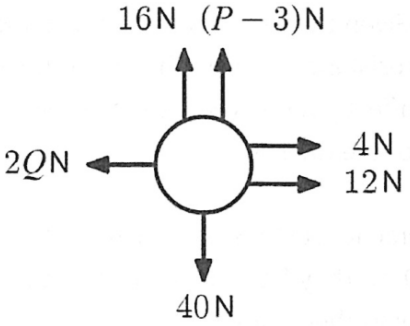


Question 3. Find the value of P and Q such that the object is at rest.

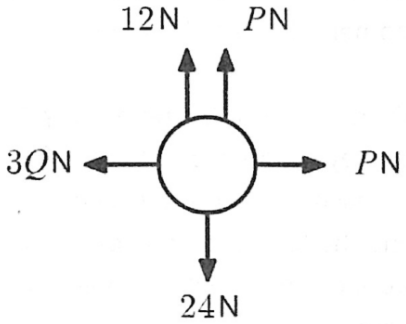
a)



b)

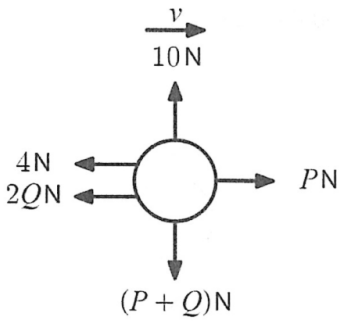


c)

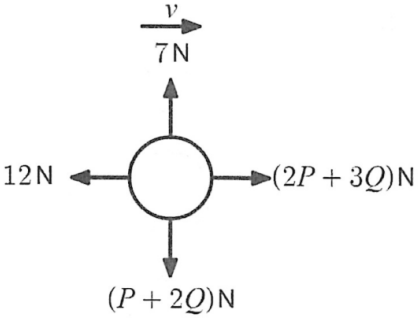


Question 4. Each of the following diagrams shows an object acted on by a set of forces. Find the value of P and Q such that the object is moving with constant velocity.

a)



b)



Question 5. An object of weight 50 N is hanging by a string in equilibrium as shown in the diagram.



a) State the value of T

b) By using $g = 10$ ms^{-2}, find the mass of the object.

The string snaps and the object drops.

c) State the size and direction of the resultant force acting on the object.

Question 6. State the weight of an object of mass 50 kg

a) on earth given $g = 9.81$ ms^{-2}.

b) on the moon given $g = 1.63$ ms^{-2}.

Question 7. An object has weight 4000 N on Earth where $g = 9.8$ ms^{-2}. Find the weight of the same object on the moon where the acceleration due to gravity is 1.6 ms^{-2}.

Question 8. An object of mass 0.4 kg is being held in equilibrium by an upwards force of P N.
Using $g = 9.8$ ms^{-2}

a) Find the weight of the object.

b) Hence, state the value of P

The upwards force is removed.

c) State the size and direction of the resultant force acting on the object.

Question 9. A small body is hanging in equilibrium at the end of a vertical string.

a) Draw a force diagram to represent the forces acting on the body.

Find, using $g = 9.8$ ms^{-2}, the

b) weight of the body if the mass is 2 kg.

c) mass of the body if its weight is 34.3 N.

d) tension in the string if the mass is 0.5 kg.

e) tension in the string if the weight of the body is 14.7 N.

Question 10. A box is at rest on a horizontal table.

a) Draw a force diagram to represent the forces acting on the box.

Find, using $g = 10$ ms^{-2}, the

b) weight of the box if the box is 1.5 kg.

c) mass of the box if its weight is 60 N.

d) mass of the box if the normal reaction of the table on the box is 5 N.

e) normal reaction of the table on the box if the mass of the box is 1.2 kg.

f) normal reaction of the table on the box if the weight of the body is 25 N.

Question 11. A book of mass 400 g is being pushed along with a force of 2 N on a smooth horizontal table. Draw a force diagram to represent the forces acting on the book.